

KNF Nutrient Program*

Updated 8/10/21

***NOTE: The following content is for a soil medium. Please see the section titled “Growing Medium Adjustments” if you are using a different medium.**

Materials Needed

1. [Humic Acid \(powder\)](#) **KNF Website**
 - i. [Humic acid \(powder\)](#) **Amazon**
 - b. At least 64%
 - c. 3+ Fulvic
2. [Soybean amino acid](#) **KNF Website**
 - i. Soybean amino acid
 - b. enzymatic/Hydrolysate form
3. [Solution Grade Langbeinite](#) **KNF Website**
 - i. [Solution Grade Langbeinite](#) **Paradise Superstore**
4. Raw light brown sugar (a lot)
5. [Bloom Ferment \(all organic\)](#) - DIY
 - a. Mango - 7 lbs (should be +2 lbs more than all other fruit)
 - b. Papaya - 5 lbs
 - c. Apple - 5 lbs
 - d. Carrot - 5 lbs
 - e. Zucchini - 5 lbs
 - f. Squash (any) - 5 lbs
 - g. Large bundle of Banana -
 - h. Large bundle of Beets (bulb, not stem)
 - i. Small watermelon
 - j. Habaneros - 2 lbs
6. [Water-Soluble Calcium](#) - DIY
 - a. Brown rice vinegar
 - b. [Oyster shell flour](#) **Organically Done**
7. [Fermented Amino Acid](#) - DIY
 - a. **Fatty** Fish (ex. Salmon bellies, tilapia) - 5 to 10 lbs
 - b. **Super Ripe** Pineapple (for every 10 lbs -- if making 5 lbs then use half lb)
 - i. Buy a pineapple and leave out for a few days to make sure it is ripe enough
8. [Soil](#) (or an inert soil)- NO NUTRIENTS ADDED

Equipment Needed

1. Food Grade 5 gallon buckets

2. Masticating Juicer
3. Blender
4. Large colander/strainer and paint strainer bags
5. Storage containers (preferably glass)

Schedule

Stage →	Veg	Veg	Veg	Veg	Bloom	Bloom	Bloom	Bloom	Bloom	Bloom	Bloom	Bloom	Bloom
Input ↓	Wk 1	Wk 2	Wk 3	Wk 4	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8	Wk 9
Humic ¼ TSP/ 5gal	X	X	X	X	X	X	X	X	X	X	X	X	X
Soybean* 2 TBSP/5gal	X	X	X	X	X	X	X	X	X	X	X		
FAA 3ml/gal		X		X		X		X		X		X	
BF 3-5ml/gal					X	X	X	X	X	X	X	X	X
WSC** 3ml/gal							X	X	X	X	X		
LSG** 1 TBSP/5gal							X	X	X	X	X		
RB 3ml/gal					X	X	X	X	X	X	X	X	X

* **NOTE:** 2 TBSP of Soybean is a STARTING dose. You can increase the amount by 1 TBSP increments up to 4 TBSP if you see any problems with leaves (i.e. not a dark lush green) during veg state. Once soybean is increased, maintain the same amount throughout your grow.

** **NOTE:** If you are growing indoors under LED lighting using CO2, use WSC and LSG beginning WEEK ONE of bloom.

Measurements

Input	Stage	Frequency	Amount	Notes
Humic	Full plant life	Every watering	¼ tsp per 5 gal	If stems are purple, up to 1 tsp until problem resolves, then resume baseline
Soybean Amino Acid	Full plant life	Every week	2 TBSP per 5 gal STARTING	If leaves are not satisfactory, up to 4 TBSP and maintain that amount
Fermented Amino Acid	Full plant life	Every 2 weeks	3ml per 1 gal	OPTIONAL - Use to recharge soil
Bloom Ferment	Bloom	Every week	3 to 5ml per gal	
Root Beer	Bloom	Every week	3ml per 1 gal	
Water Soluble Calcium	Week #3 of Bloom**	Every week	3ml per 1 gal	If you grow indoors under LED lighting using CO2, begin WSC during Week #1 of Bloom
Langbeinite Solution Grade**	Week #3 of Bloom	Every week	1 TBSP per 5 gal	If you grow indoors under LED lighting using CO2, begin WSC during Week #1 of Bloom
Fermented Plant Extract	As needed	As needed	3ml per 1 gal	

Instructions

(BF) Bloom Ferment

Materials	Equipment	Measurements/Amounts
mango, papaya, apple, carrot, zucchini, squash, banana, beet, watermelon, habaneros, brown sugar	juicer Blender 5-gal bucket	3-5 ml per 1 gallon of water Use every watering during bloom to harvest

1. Cut up all fruit
2. Juice the carrots, apples, and beets first.
3. Use some of the juice to blend other ingredients in blender on low (think chunky salsa)
4. Add all pulp and leftover juice to a 5 gal bucket and add brown sugar at a 1:1 ratio
5. Mix together thoroughly
6. Add habaneros and mix in - be careful!
7. Let ferment for 15 days, mixing every 3 days
8. Watch the video in Discord channel for best straining technique. Store in a glass jar with a breathable lid.

(WSC) Water-Soluble Calcium

Materials	Equipment	Measurements/Amounts
Oyster shell flour Brown rice vinegar	Large glass jar 5-gal bucket	3 ml per 1 gallon of water Begin using Week 3 of bloom, use weekly** (see notes above)

1. Place the jar in the 5-gal bucket (this is to reduce mess)
2. Fill $\frac{1}{4}$ of the jar with oyster shell flour
3. Fill the rest of the jar with brown rice vinegar
4. Mix together thoroughly -- be careful, it will have a violent reaction
5. Let the reaction happen. When the bubbling stops, the WSC is ready. There will still be a lot of oyster shell flour at the bottom -- that is normal.

6. To strain: Use a 1-gal paint strainer bag. Pour the mixture into the strainer bag over the final holding container or a bowl and transfer to the final holding container. Store with a sealed lid.

(RB) Root Beer

Materials	Equipment	Measurements/Amounts
Root ball of used plants Brown sugar water	5-gal bucket	3 ml per 1 gallon of water Use weekly during bloom

1. Pull root balls out of their containers and shake until dirt is not loosely falling (no need to remove ALL dirt)
2. Put 4-5 root balls in the 5-gal bucket
3. Fill bucket with water and add two cups of brown sugar. Mix.
4. Every 2 days, skim the excess dirt from the top
5. On day 5, add 4 cups of brown sugar and mix.
6. Let ferment for 15 days
7. After 15 days, strain and collect the liquid
8. 1-2 weeks after collection, add some more brown sugar to help with odor.
9. To strain: Use a large strainer/colander placed on top of a clean 5-gal bucket. Pour the mixture into the strainer, then transfer the liquid to the final holding container. Store with a breathable lid.

(FPE) Fermented Plant Extract - reduce stress as needed, ease transplanting

Materials	Equipment	Measurements/Amounts
Cannabis leaves Brown sugar	Two 5-gal buckets Rocks (or other weight)	3-5 ml per 1 gallon of water Use as needed

1. Collect cannabis leaves and shoots
2. Add two handfuls of leaves and two handfuls of brown sugar to a 5-gal bucket
3. Massage sugar and leaves together until the leaves are wet (should look like cooked spinach)

4. Repeat steps 2-3 until the 5-gal bucket is full
5. Put the second 5-gal bucket on top of the cannabis leaves mixture and add some kind of weight
6. Let ferment for 15 days, then strain and collect liquid.
 - a. Use this liquid 3ml/gal
7. Leave the brick of cannabis leaves in the bucket and add 8 cups of water.
8. After hydrating for 1 day, ring out the brick of leaves and collect liquid. Discard leaves.
 - a. Use this liquid 5ml/gal
9. To strain: Use a paint strainer bag and put it on a 1 or 5 gal bucket (size depends on amount made). Pour the leaves and mixture into the strainer bag. Release the strainer bag from the edges of the bucket. Squeeze the end of the strainer bag until liquid stops coming out. Store with a breathable lid.

(FAA) Fermented Amino Acid - OPTIONAL (recharges soil)

Materials	Equipment	Measurements/Amounts
Fatty Fish Brown sugar Very ripe pineapple	Blender 5-gal bucket	3 ml per 1 gallon of water Use every two weeks from start to finish

1. Roughly cut up a minimum of 5 lbs of any fish (typically tilapia or salmon) and put in a 5-gal bucket.
2. Add brown sugar in a 1:1 ratio
3. Completely coat fish in the brown sugar
4. Roughly cut up a pineapple and blend until pureed.
 - a. You will need 1 pineapple for every 10 lbs of fish ($\frac{1}{2}$ pineapple for 5 lbs)
5. Add the pineapple puree to the fish/sugar and mix very well
6. Press the mixture down until the top is flattened
7. Top with brown sugar (called sugar cap) until you can no longer see the mixture
8. Let it ferment for 30 days (no need to add a second sugar cap)
9. After fermenting, strain the mixture to collect the liquid.
10. To strain: Use a large strainer/colander and put it on a 5 gal bucket. Pour the mixture into the strainer and move the contents around until liquid stops coming out of the bottom. Store with a breathable lid.

Growing Medium Adjustments

- If using soilless or coco:
 - Feeding any nutrient denoted as *weekly* in the **Schedule** section may be used EVERY watering at a reduced rate based on plant reaction.

Troubleshooting

- Exactly how much sugar do you add into the bloom ferment?
 - You can never add too much sugar! Add it until the consistency is a little syrup-y. You can even add sugar after you've strained it until sugar cannot dissolve anymore.
- There is mold growing on top of my ferments (before or after straining)!
 - White/blue/green mold is fine. Mold is most likely to occur on the bloom ferment if you are not stirring frequently enough. Mold is most likely to occur on the FAA if pieces of fish are sticking above the sugar cap. Remove the mold and carry on. If the mold is black, you have a problem and need to contact KNF Guy to troubleshoot.
- What does a breathable lid mean?
 - That means something with holes. Cheese cloth is a good alternative if your lids are unable to be perforated.

WARNINGS

1. **Bloom Ferment:** You CAN burn plants using this, so be diligent. Burns will be isolated and may be difficult to find.

PICTURES



This is what an ideal cannabis leaf should look like. It has been provided enough Soybean to develop and maintain this rich green color.